

$$??h\Omega$$

$$(1) \quad \{ \begin{array}{l} M_{\alpha\beta}-Q_{\alpha}=0 \text{in } \Omega \\ Q_{\alpha,\alpha}+\bar{q}=0 \text{in } \Omega \\ Q_{\alpha}n_{\alpha}=\bar{Q} \text{on } \Gamma_Q \\ M_{\alpha\beta}n_{\beta}=\bar{M}_{\alpha} \text{on } \Gamma_M \\ \varphi_{\alpha}=\bar{\varphi}_{\alpha} \text{on } \Gamma_{\varphi} \\ w=\bar{w} \text{on } \Gamma_w \end{array}$$

$$(2) \quad M_{\alpha\beta}=-\frac{h^3}{12}D_{\alpha\beta\gamma\eta}\kappa_{\gamma\eta}=\frac{h^3}{12}D_{\alpha\beta\gamma\eta}\varphi_{\gamma,\eta}$$

$$(3) \quad Q_{\alpha}=k\frac{Eh}{2(1+\nu)}\gamma_{\beta}=kGh(-\varphi_{\beta}+w_{,\beta})$$

$$(4) \quad \begin{array}{l} \kappa_{\alpha\beta}=-\varphi_{\alpha,\beta}\gamma_{\alpha}=-\varphi_{\alpha}+w_{,\alpha} \\ D_{\alpha\beta\gamma\eta} \end{array}$$

$$(5) \quad D_{\alpha\beta\gamma\eta}=\frac{E}{1-\nu^2}(\nu\delta_{\alpha\beta}\delta_{\gamma\eta}+\frac{1}{2}(1-\nu)(\delta_{\alpha\gamma}\delta_{\beta\eta}+\delta_{\alpha\eta}\delta_{\beta\gamma}))$$

$$(6) \quad \begin{array}{l} m indlin \Pi(w,\varphi)=\\ \frac{1}{2}\int_{\Omega}\kappa_{\alpha\beta}M_{\alpha\beta}d\Omega+\\ \frac{1}{2}\int_{\Omega}\gamma_{\alpha}Q_{\alpha}d\Omega\\ +\\ \int_{\Gamma_M}\varphi_{\alpha}\bar{M}_{\alpha}d\Gamma-\\ \int_{\Gamma_Q}w\bar{Q}d\Gamma-\\ \int_{\Omega}w\bar{q}d\Omega\\ e^{energy}(\varphi_{\alpha})\in V \end{array}$$

$$(7) \quad \int_{\Omega}\delta\kappa_{\alpha\beta}M_{\alpha\beta}d\Omega+\int_{\Omega}\delta\gamma_{\alpha}Q_{\alpha}d\Omega=-\int_{\Gamma_M}\delta\varphi_{\alpha}\bar{M}_{\alpha}d\Gamma+\int_{\Gamma_Q}\delta w\bar{Q}d\Gamma+\int_{\Omega}\delta w\bar{q}d\Omega,\forall(\delta w,\delta\varphi_{\alpha})\in V$$

$$(8) \quad \frac{\Omega\{x_I\}_{I=1}^{n_u}n_u}{w\delta w\varphi_{\alpha}\delta\varphi_{\alpha}x_I}$$

$$(9) \quad w^h(x)=\sum_{I=1}^{n_u}N_I(x)w_I,\delta w^h(x)=\sum_{I=1}^{n_u}N_I(x)\delta w_I$$

$$(10) \quad \varphi_{\alpha}^h(x)=\sum_{I=1}^{n_u}N_I(x)\varphi_{\alpha I},\delta\varphi^h(x)=\sum_{I=1}^{n_u}N_I(x)\delta\varphi_{\alpha I}$$

$$(11) \quad \kappa^h=\kappa_{11}^h\kappa_{22}^h2\kappa_{12}^h=-\sum_{I=1}^{n_u}0N_{I,1}000N_{I,2}0N_{I,1}w_I\varphi_{1I}\varphi_{2I}=-\sum_{I=1}^{n_u}B_I^bd_I$$

$$(12) \quad \gamma^h=\gamma_1^h\gamma_2^h=\sum_{I=1}^{n_u}N_{I,1}N_I0N_{I,2}0N_Iw_I\varphi_{1I}\varphi_{2I}=\sum_{I=1}^{n_u}B_I^sd_I$$

$$(13) \quad \delta\kappa^h=-\sum_{I=1}^{n_u}B_I^b\delta d_I,\delta\gamma^h=\sum_{I=1}^{n_u}B_I^s\delta d_I$$

$$(14) \quad \begin{array}{l} hweak_penalty m indlin^h, \varphi_{\alpha}^h) \in \\ V_h \end{array}$$

$$(15) \quad \int_{\Omega}\delta\kappa_{\alpha\beta}^hM_{\alpha\beta}d\Omega+\int_{\Omega}\delta\gamma_{\alpha}^hQ_{\alpha}d\Omega=-\int_{\Gamma_M}\delta\varphi_{\alpha}^h\bar{M}_{\alpha}d\Gamma+\int_{\Gamma_Q}\delta w^h\bar{Q}d\Gamma+\int_{\Omega}\delta w^h\bar{q}d\Omega,\forall(\delta w^h,\delta\varphi_{\alpha}^h)\in V_h$$

$$(16) \quad \kappa^h\gamma^h$$